

# Current Diagnostic Approach and Therapy Selection for Thyroid Nodules



# Benign or Malignant ??



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M

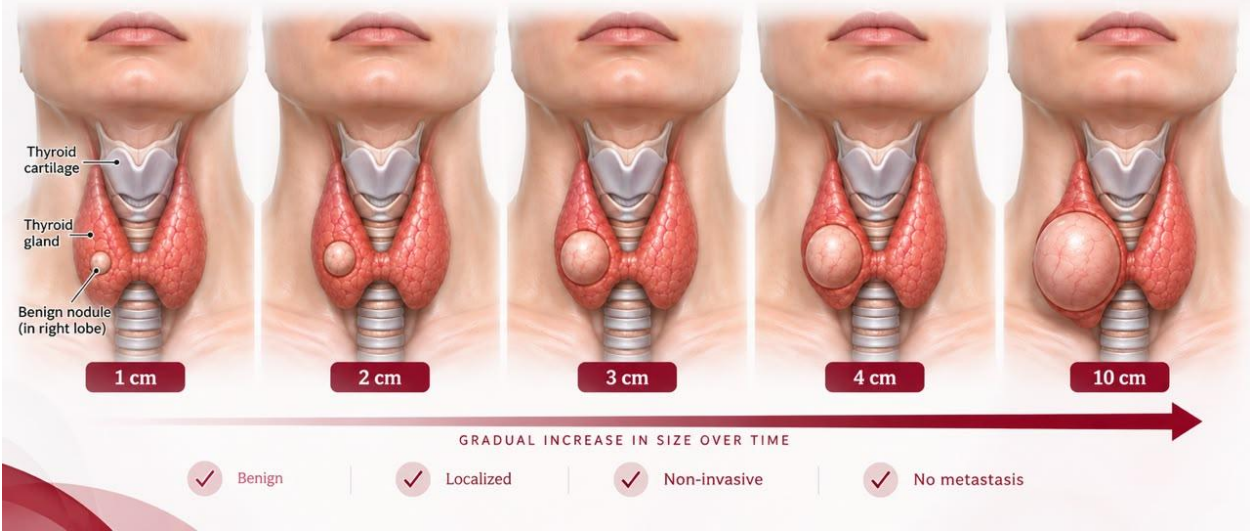


**BAMEDICALE**  
EST. 2024

EDUCATION | DISCOVERY | BETTER CARE  
FOR A HEALTHIER TOMORROW

# Benign Thyroid Nodule – Early Progression

Same anatomy. Gradual increase in nodule size (benign, localized, non-invasive).



✓ Benign

✓ Localized

✓ Non-invasive

✓ No metastasis

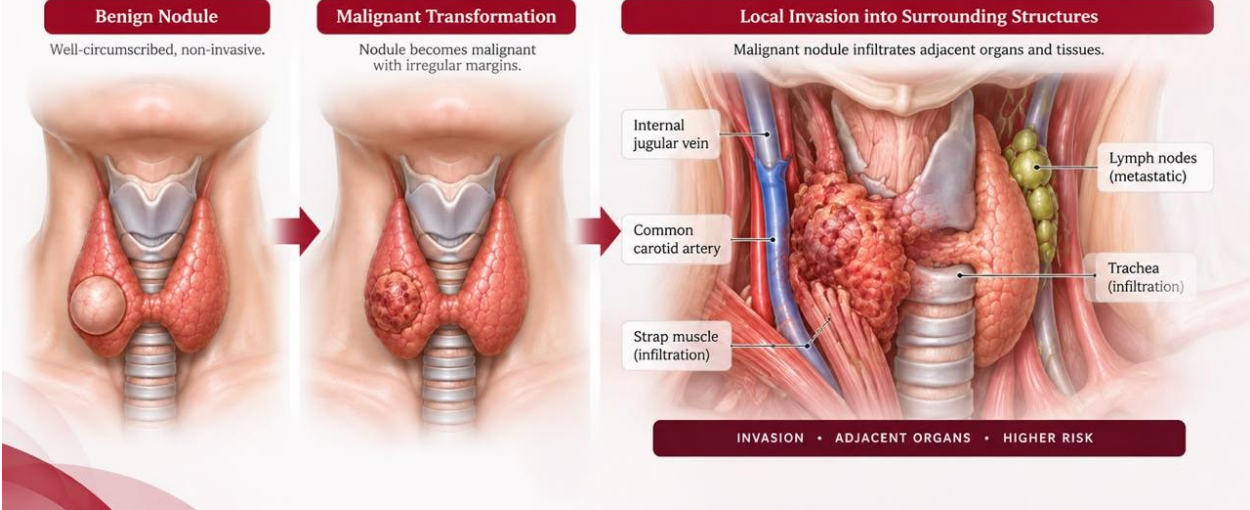
# From Benign to Malignant – Local Invasion

Progression from a benign thyroid nodule to malignancy, with invasion into surrounding structures.

**Benign Nodule**  
Well-circumscribed, non-invasive.

**Malignant Transformation**  
Nodule becomes malignant with irregular margins.

**Local Invasion into Surrounding Structures**  
Malignant nodule infiltrates adjacent organs and tissues.



Internal jugular vein

Common carotid artery

Strap muscle (infiltration)

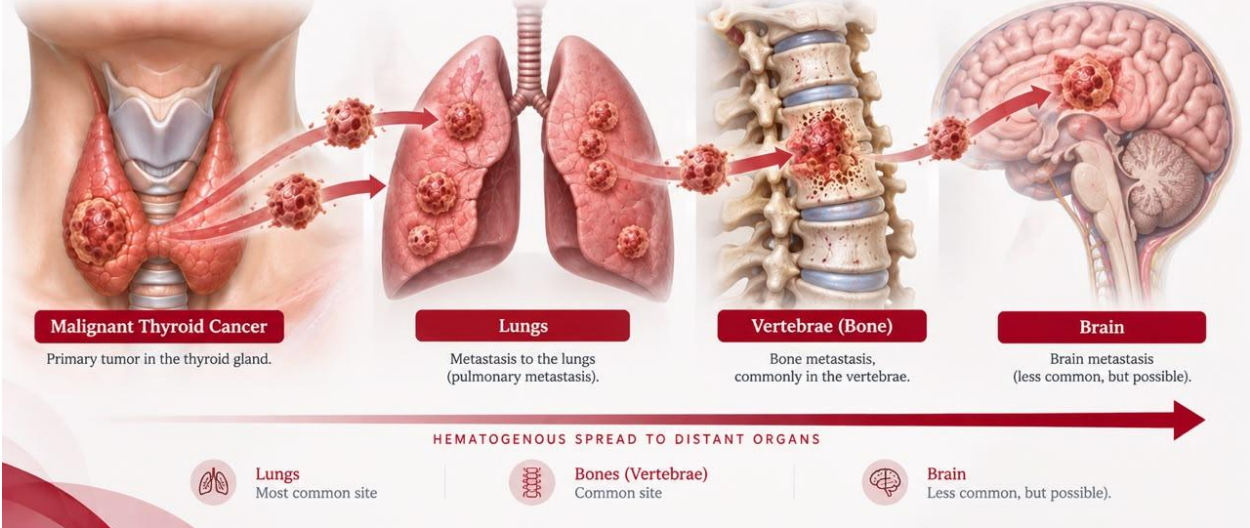
Lymph nodes (metastatic)

Trachea (infiltration)

INVASION • ADJACENT ORGANS • HIGHER RISK

# Distant Metastasis – Spread Beyond the Neck

Malignant thyroid cells can spread to distant organs, most commonly the lungs, bones (vertebrae), and brain.



**Malignant Thyroid Cancer**  
Primary tumor in the thyroid gland.

**Lungs**  
Metastasis to the lungs (pulmonary metastasis).

**Vertebrae (Bone)**  
Bone metastasis, commonly in the vertebrae.

**Brain**  
Brain metastasis (less common, but possible).

HEMATOGENOUS SPREAD TO DISTANT ORGANS

 **Lungs**  
Most common site

 **Bones (Vertebrae)**  
Common site

 **Brain**  
Less common, but possible.

Incidence, Both sexes, in 2024

| Cancer                                     | ICD Code   | Number     | ASR<br>(World) | Crude<br>Rate | Cum.<br>risk 74 |
|--------------------------------------------|------------|------------|----------------|---------------|-----------------|
| All cancers excl. non-melanoma skin cancer | C00-97/C44 | 19 498 127 | 185.6          | 240.2         | 19.1            |
| Trachea, bronchus and lung                 | C33-34     | 2 637 005  | 23.9           | 32.5          | 2.9             |
| Breast                                     | C50        | 2 434 087  | 47.8           | 60.3          | 5.2             |
| Colorectum                                 | C18-C21    | 2 041 007  | 18.6           | 25.2          | 2.1             |
| Prostate                                   | C61        | 1 546 112  | 29.3           | 37.9          | 3.6             |
| Stomach                                    | C16        | 980 286    | 8.9            | 12.1          | 1.0             |
| Thyroid                                    | C73        | 959 281    | 10.3           | 11.8          | 1.0             |
| Liver and intrahepatic bile ducts          | C22        | 843 045    | 7.9            | 10.4          | 0.93            |
| Other specified cancers                    | OTH        | 732 099    | 7.3            | 9.0           | 0.74            |
| Bladder                                    | C67        | 635 264    | 5.5            | 7.8           | 0.62            |
| Cervix uteri                               | C53        | 604 196    | 12.4           | 15.0          | 1.3             |

World no.6

| Cancer                                     |
|--------------------------------------------|
| All cancers excl. non-melanoma skin cancer |
| Trachea, bronchus and lung                 |
| Breast                                     |
| Colorectum                                 |
| Thyroid                                    |
| Stomach                                    |
| Liver and intrahepatic bile ducts          |
| Prostate                                   |
| Other specified cancers                    |
| Oesophagus                                 |
| Cervix uteri                               |

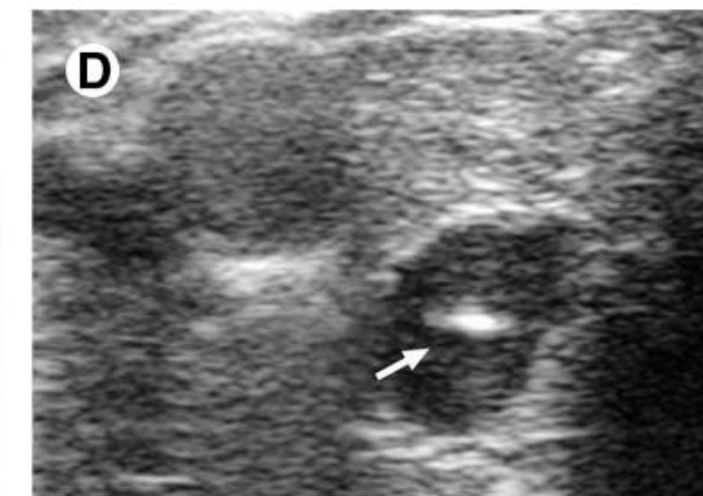
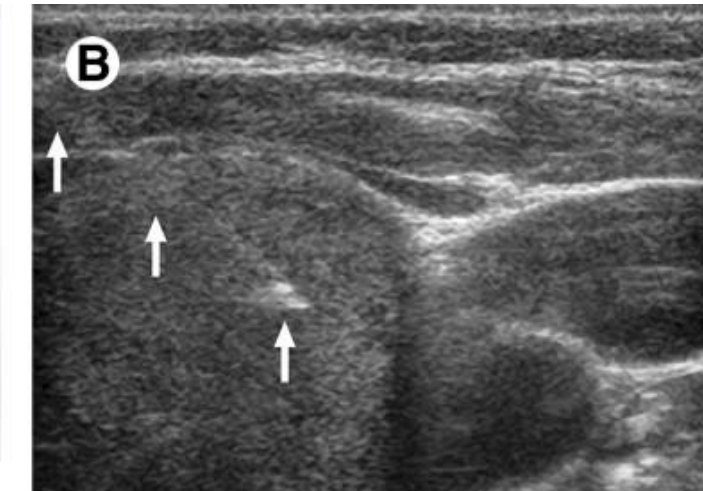
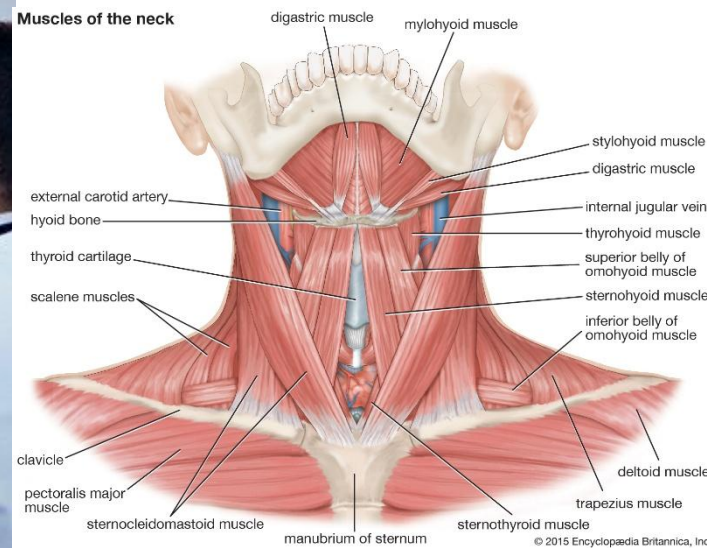
Asia no.4

Dharmais no.4

|                                     |       |
|-------------------------------------|-------|
| PAYUDARA (C50)                      | 29,12 |
| TRAKHEA, BRONKUS, DAN PARU (C33-34) | 6,76  |
| SERVIKS UTERI (C53)                 | 6,61  |
| TIROID (C73)                        | 6,05  |
| LEUKEMIA (C91-95)                   | 5,29  |
| NASOFARING (C11)                    | 4,12  |
| LIMFOMA NON HODGKIN (C82-86,96)     | 3,64  |
| REKTUM (C19-20)                     | 3,50  |
| KOLON (C18)                         | 3,42  |
| OVARIUM (C56)                       | 3,02  |

# Diagnostic

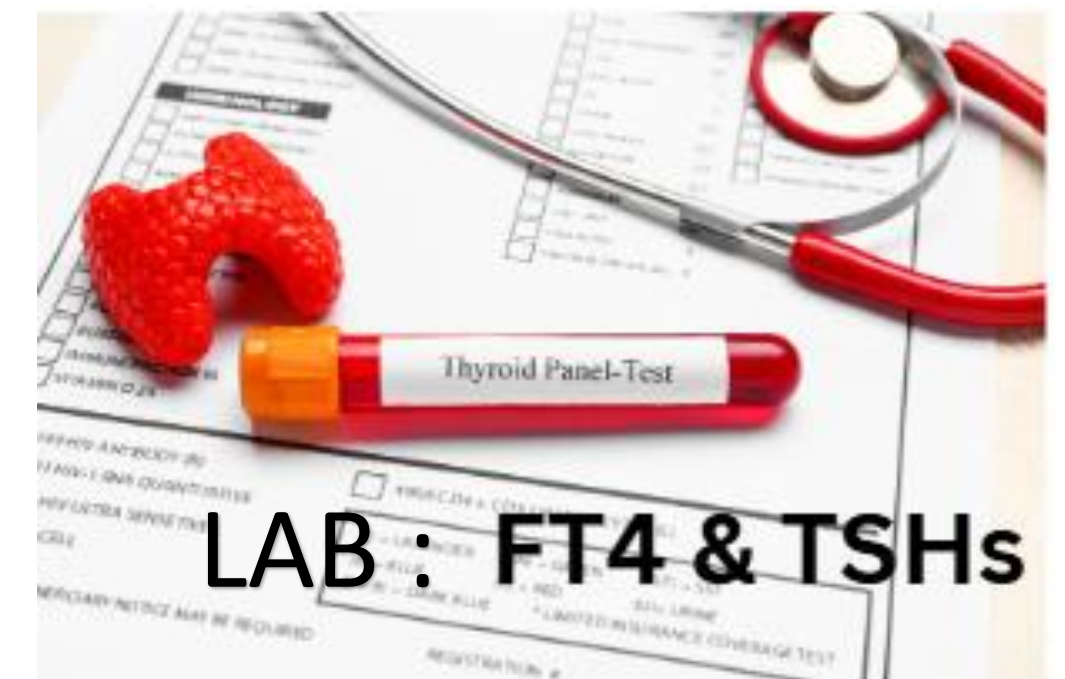
## Physical Examination



FNAB →  
Bethesda



Imaging :  
Ultrasound  
→ TIRADS



LAB : FT4 & TSHs

# NCCN Guidelines Version 2.2026

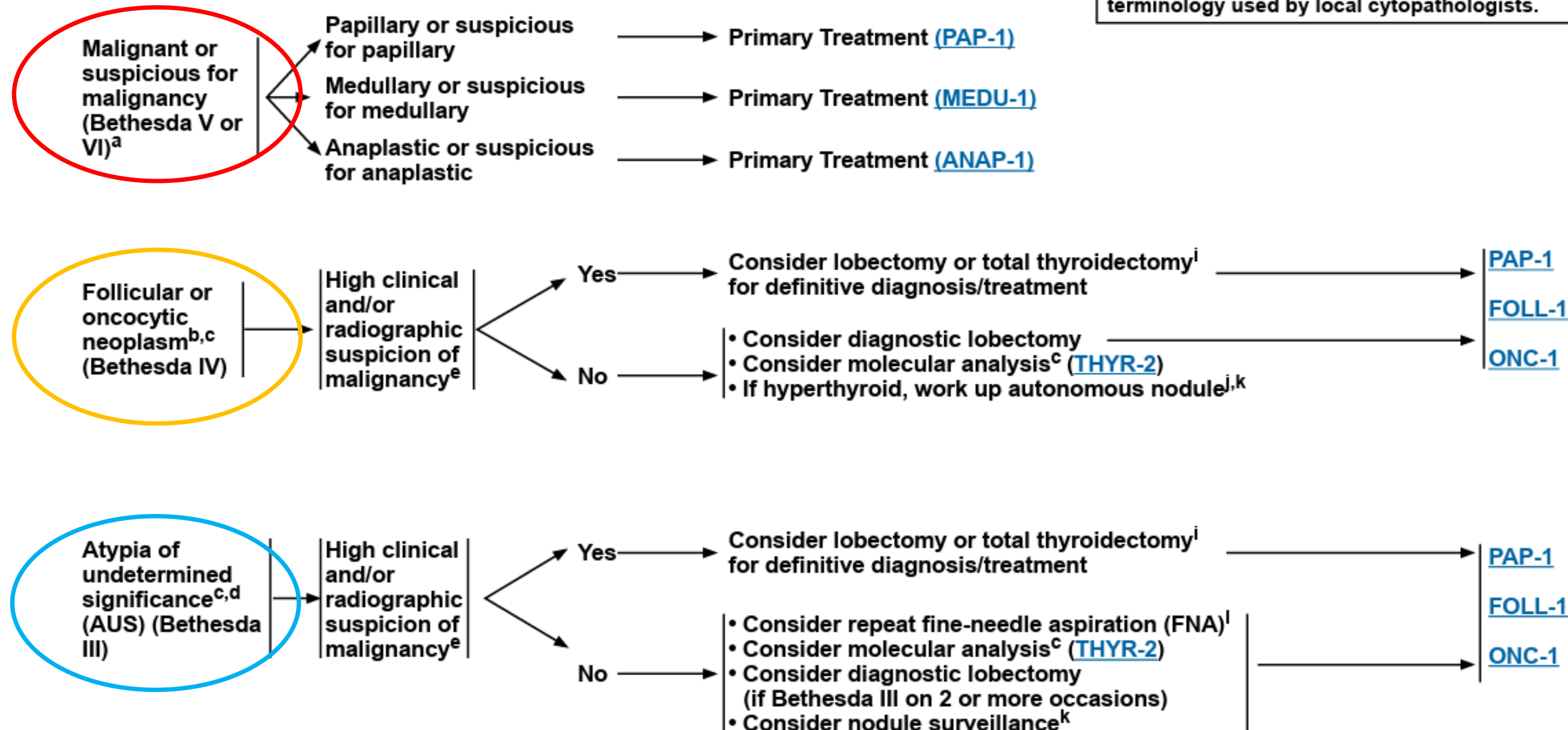
## Thyroid Carcinoma

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### FNA RESULTS

### TREATMENT<sup>f,g,h</sup>

Diagnostic categories for FNA results reflect NCI state of the science conference, the [Bethesda Classification](#) (Ali SZ, Baloch ZW, Cochand-Priollet B, et al. Thyroid 2023;33:1039-1044). Cytology reports should be interpreted in light of terminology used by local cytopathologists.



Bethesda III & IV → need to  
*Molecular Analysis* to get  
diagnosis Benign or Malignant

## Risk of Malignancy

- Bethesda VI Estimated = 97%
- Bethesda V = 74%
- Bethesda IV = 30%



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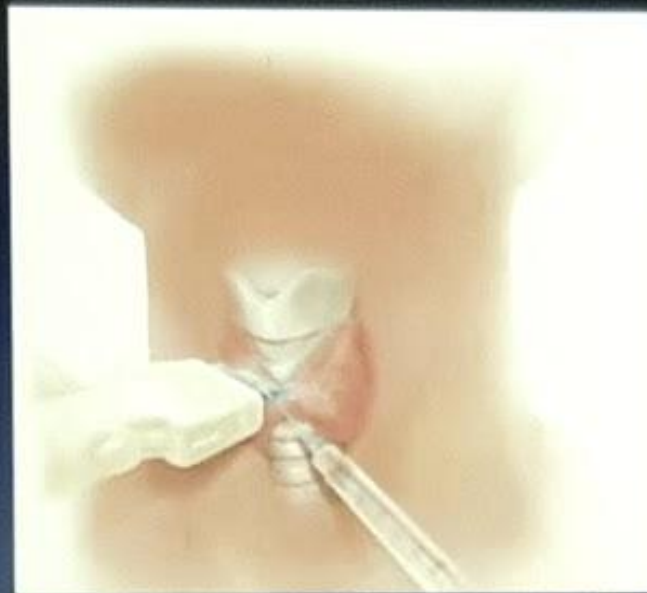
## NCCN Guidelines Version 2.2026 Thyroid Carcinoma

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### FOOTNOTES

- <sup>a</sup> Bethesda V Estimated risk of malignancy, inclusive of noninvasive follicular thyroid neoplasm with papillary-like nuclear features (NIFTP), mean 74% (range, 67%–83%); Bethesda VI estimated risk of malignancy, inclusive of NIFTP, mean 97% (range, 97%–100%); Ali SZ, et al. *Thyroid* 2023;33:1039-1044.
- <sup>b</sup> Bethesda v3 terminology for Bethesda IV is follicular neoplasm or oncocytic follicular neoplasm, and the estimated risk of malignancy, inclusive of NIFTP, is mean 30% (range, 23%–34%).
- <sup>c</sup> The diagnosis of follicular carcinoma or oncocytic carcinoma requires evidence of either vascular or capsular invasion, which cannot be determined by FNA. Molecular analysis may be useful to allow reclassification of follicular lesions (ie, follicular neoplasm, AUS) as either more or less likely to be benign or malignant based on the molecular profile. If molecular testing suggests papillary thyroid carcinoma (PTC), especially in the case of *BRAF* V600E, see [PAP-1](#). Given the challenges of cytology to explicitly diagnose medullary thyroid cancer (MTC) in limited samples, molecular tests may be used to identify them. If molecular testing, in conjunction with clinical and ultrasound features, predicts a risk of malignancy comparable to the risk of malignancy seen with a benign FNA cytology (approximately ≤4%), consider nodule surveillance. Molecular analysis results should be interpreted with caution and in the context of clinical, radiographic, and cytologic features of each individual patient. If molecular analysis is technically inadequate or not done, then repeat FNA for molecular testing if available.
- <sup>d</sup> Estimated risk of malignancy is mean 22% (range, 13%–30%) inclusive of NIFTP.
- <sup>e</sup> Based on rapid growth of nodule, imaging, physical examination, age, clinical history of radiation, and family history.
- <sup>f</sup> The order of the treatment options does not indicate preference.
- <sup>g</sup> Refer to the NCCN Distress Thermometer and Problem List, which includes social determinants of health. See [NCCN Guidelines for Distress Management \(DIS-A\)](#).
- <sup>h</sup> Clinical risk factors, sonographic patterns, reported risk of malignancy based on molecular test, and patient preference can help determine whether nodule surveillance or surgery is appropriate.
- <sup>i</sup> Total thyroidectomy may be considered for history of radiation exposure or contralateral lobe lesions.
- <sup>j</sup> [Principles of Active Surveillance for Low-Risk Papillary Thyroid Cancer \(THYR-D\)](#).
- <sup>k</sup> Follow American Thyroid Association (ATA) guidelines for nodule surveillance (<https://pmc.ncbi.nlm.nih.gov/articles/PMC4739132/pdf/thy.2015.0020.pdf>).
- <sup>l</sup> Consider second opinion pathology at high-volume center.

# Molecular Testing



Data from Roberto Dina  
ICNHST 2014

\*10-25% in other series

|            | Non-diagnostic | Benign              | Indeterminate      |        | Suspicious | Malignant |
|------------|----------------|---------------------|--------------------|--------|------------|-----------|
|            | Thy 1          | Thy 2               | Thy 3a             | Thy 3f | Thy 4      | Thy 5     |
| Incidence  | 15%            | 70%                 | 8%*                |        | 2%         | 5%        |
| Management | Repeat FNA     | Clinical management | Diagnostic surgery |        | Surgery    | Surgery   |

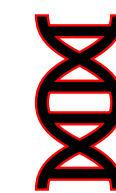
74%  
Benign

Surgery was  
unnecessary

26%  
Malignant

Need definitive  
surgery

- ThyroSeq v3 is also based on next-generation sequencing of DNA and RNA.
- Expanded to analyze 112 genes,
- Providing information on >12,000 mutation hotspots and >150 gene fusion
- 1st time launched on april 2007 in Pittsburg University



**Dharmais begins  
by NGS in trial 2026**

(Nikiforova MN et al. 2018).

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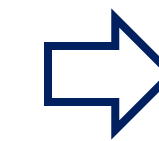
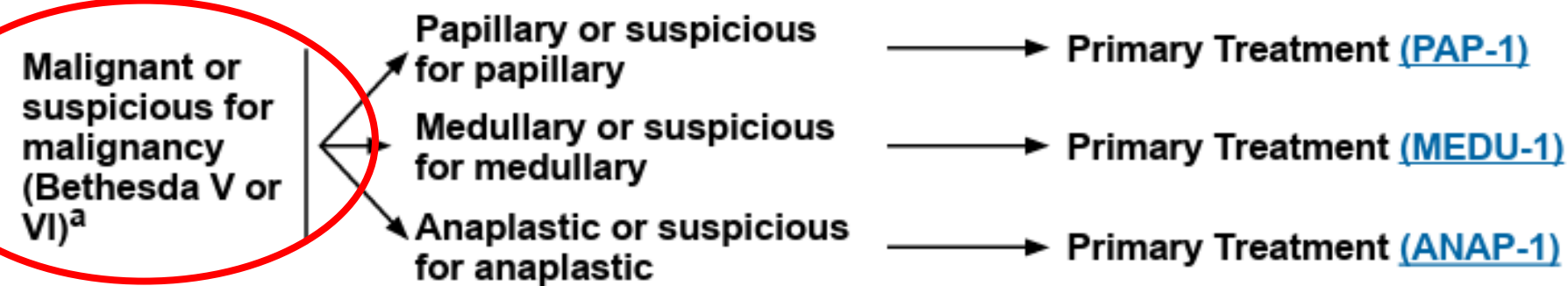
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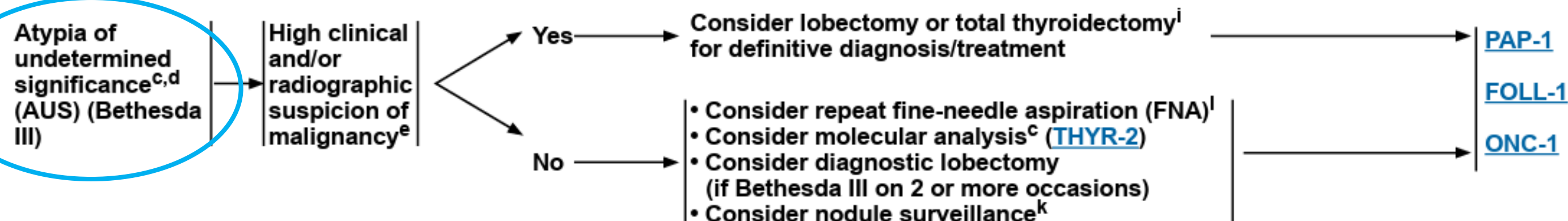
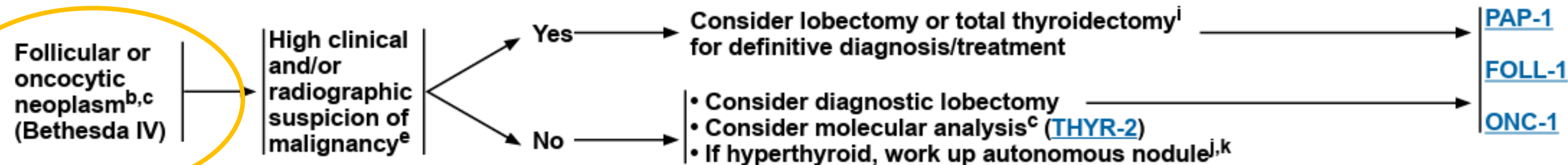
### FNA RESULTS

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**Surgery : Open / Endoscopic / Robotic**

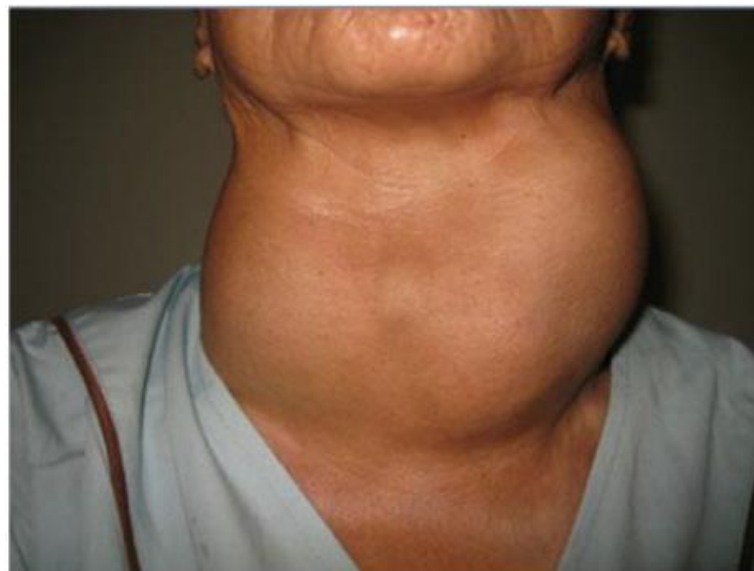


**Ablative Treatment**



# Benign Tiroid Nodule

- Conservative : < 10 mm, TIRADS 1-2
- Surgery : Open, Endoscopic or Robotic
- Non-Surgery (Ablative therapy) : Percutaneous Ethanol Ablation, Microwave, RFA & HIFU



Before RFA



18 months after RFA

# Percutaneous Ethanol Ablation



## CASE REPORT

INDONESIAN  
JOURNAL OF CANCER

## Percutaneous Ethanol Ablation for Management of Cystic Thyroid Nodules: A Case Report

*Kristanto Yuli Yarso<sup>1</sup>, Monica Bellynda<sup>2</sup>*

<sup>1</sup> Division of Surgical Oncology, Department of Surgery, Dr. Moewardi General Hospital, Surakarta, Indonesia

<sup>2</sup> Resident of Surgery, Faculty of Medicine, Sebelas Maret University, Dr. Moewardi General Hospital, Surakarta, Indonesia

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*\*Corresponding author:*  
Monica Bellynda  
Resident of Surgery, Faculty of Medicine,  
Sebelas Maret University,  
Dr. Moewardi General Hospital,  
Surakarta, Indonesia  
email: monicabellynda@hotmail.com

### ABSTRACT

**Introduction:** Percutaneous Ethanol Ablation (PEA) has been recommended as the first-line treatment option for symptomatic benign cystic or predominantly cystic thyroid nodules, and it has been shown to be more effective and safer than other techniques in previous studies. Here, we present a case of a 44-year-old man with thyroid nodules who underwent PEA.

**Case Presentation:** We report the case of a 44-year-old man with a painless mass that is getting bigger on his right neck. The physical examination measured 3.0 × 3.0 × 3.0 cm, mobile on swallowing, no pain on palpation. Thyroid function tests show a euthyroid state (TSH 2.4 mIU/L, FT4 1.2 ng/dL). The ultrasound examination in January 2020 showed a colloid cyst on the right thyroid, measured 3.03 × 2.82 × 3.56 cm (TIRADS 1). Cytological examination showed the results of non-infectious cystic fluid. The patient underwent US-guided PEA using 7 ml of ethanol by the moving-shot technique. The procedure was well tolerated by the patient and no adverse events were noted. A week after the procedure, there was a visible and progressive reduction of the nodule with no complaint from the patient. A month after the procedure, the patient came back with no recurrences, complaints, or other complications.

**Conclusions:** PEA for thyroid lesions is an effective and safe method. PEA should be the first option for the treatment of pure cystic and predominantly cystic nodules. PEA is a safe procedure that is easy to repeat and to be performed on an outpatient basis.



# RFA

radiofrequency ablation



First-line  
treatment option  
for benign nodule



Review Article

## Current guidelines for the application of radiofrequency ablation for thyroid nodules: a narrative review

Michael S. Lui<sup>^</sup>, Kcpal N. Patel

Division of Endocrine Surgery, Department of Surgery, New York University Langone Health, New York, NY, USA

*Contributions:* (I) Conception and design: Both authors; (II) Administrative support: Both authors; (III) Provision of study materials or patients: Both authors; (IV) Collection and assembly of data: Both authors; (V) Data analysis and interpretation: Both authors; (VI) Manuscript writing: Both authors; (VII) Final approval of manuscript: Both authors.

*Correspondence to:* Michael S. Lui, MD. Resident Physician, Division of Endocrine Surgery, Department of Surgery, New York University Langone Health, 530 First Avenue, New York, NY 10016, USA. Email: Michael.Lui@nyulangone.org.

**Background and Objective:** Thyroid nodules are frequently incidentally found on physical exam or imaging for an unrelated work-up. Although surgery remains the gold standard for treating symptomatic benign and/or malignant thyroid nodules, radiofrequency ablation (RFA) has emerged as a minimally invasive treatment option for high risk patients and those who decline surgery. The novel application of RFA to treat thyroid disease was originally described for symptomatic, benign thyroid nodules. Since then, several studies have tried to expand its indication to treat primary and recurrent well-differentiated thyroid cancer. The high success rates and the low complication profile, has allowed for quick adoption of RFA as a treatment option for well-selected patients with benign thyroid nodules and locoregional recurrent thyroid malignancy. As such, multidisciplinary guidelines and consensus statements were developed to standardize indications, techniques, outcome measures, and follow-up to ensure the best patient care. This article summarizes the current indications and recommendations to help guide clinicians on how best to effectively and safely utilize RFA to treat thyroid disease.

**Methods:** A PubMed/MEDLINE search between 2000–2022 using a combination of “radiofrequency ablation”, “RFA”, “thyroid nodule”, and “guidelines” was conducted. The inclusion criteria were articles published in English which offered recommendations on RFA use for thyroid nodules.

**Key Content and Findings:** For symptomatic, benign thyroid nodules, RFA is effective at significantly reducing nodule volume. For large nodules, multiple RFA sessions may be needed to achieve clinically significant volume reduction. Patients undergoing RFA for autonomously functioning thyroid nodules may see symptomatic relief but success rates are variable. RFA may serve a curative or palliative role in recurrent well-differentiated thyroid cancers. There is little data describing the use of RFA for primary well-differentiated thyroid cancer >1 cm and the role of RFA for thyroid microcarcinomas remains controversial.

**Conclusion:** RFA is a safe minimally invasive technique and may be considered, in appropriate circumstances, a first-line treatment option for benign thyroid nodules. Practices adopting RFA will likely increase as more clinicians become familiar with this technique, highlighting the importance of developing standardized guidelines.

**Keywords:** Thyroid nodules; radiofrequency ablation (RFA); autonomously functioning thyroid nodules (AFTN); recurrent thyroid cancer

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## MANAGEMENT OF THYROID NODULES : HOW TO MAKE A GOOD DIAGNOSIS ?

Live Webinar, September 19<sup>th</sup> 2026

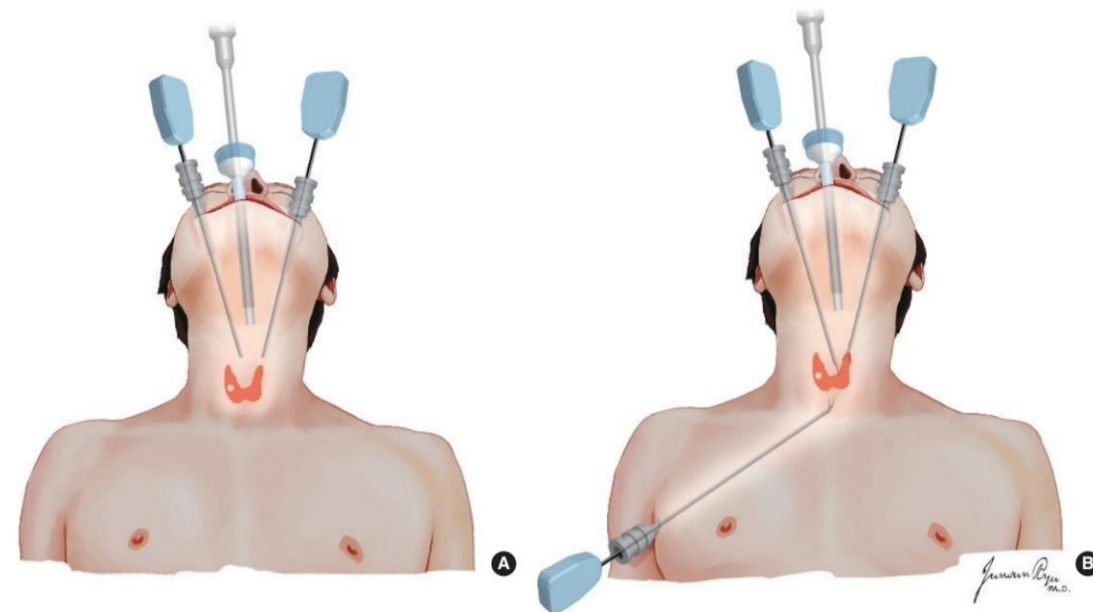
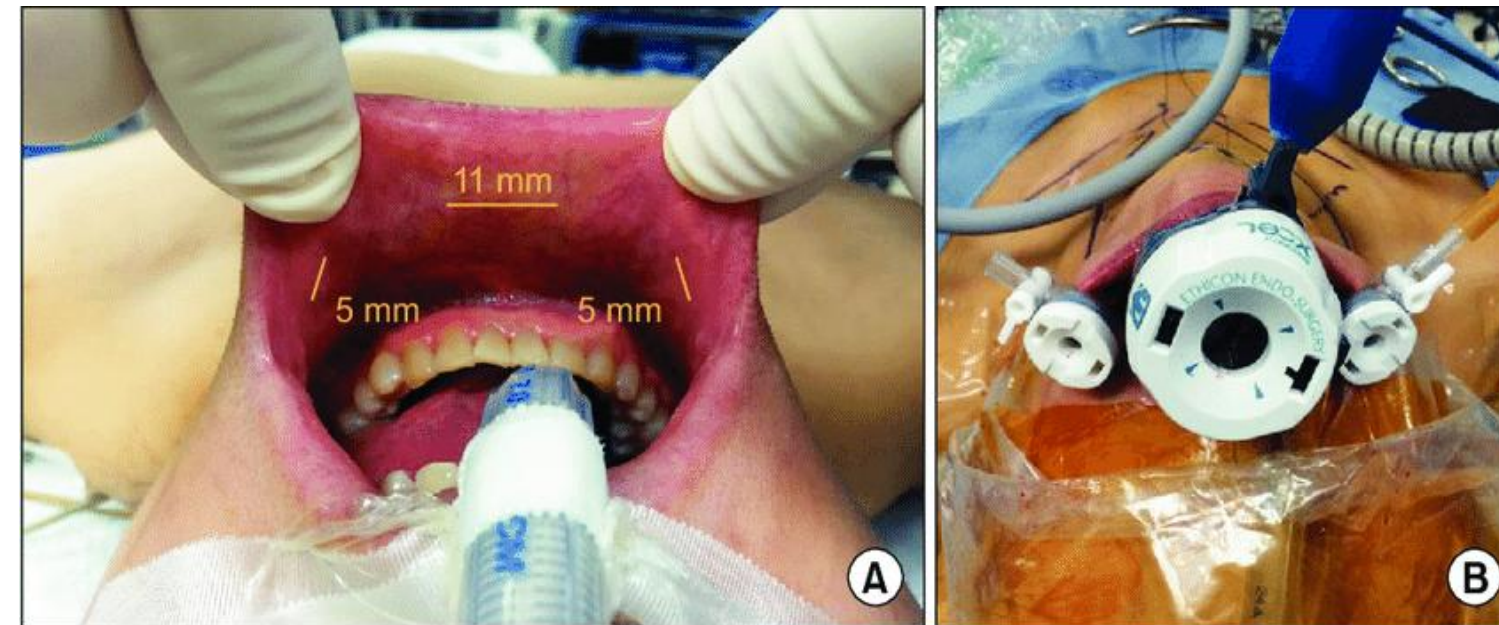
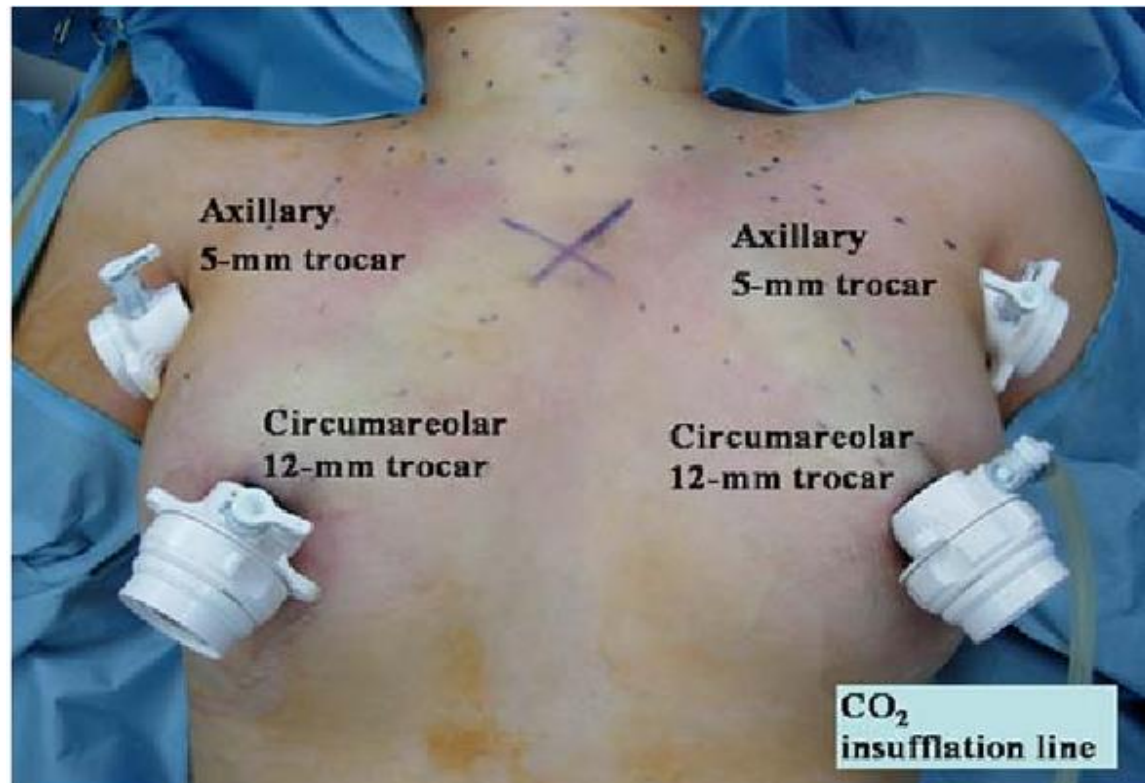


# Well Differentiated Carcinoma Thyroid

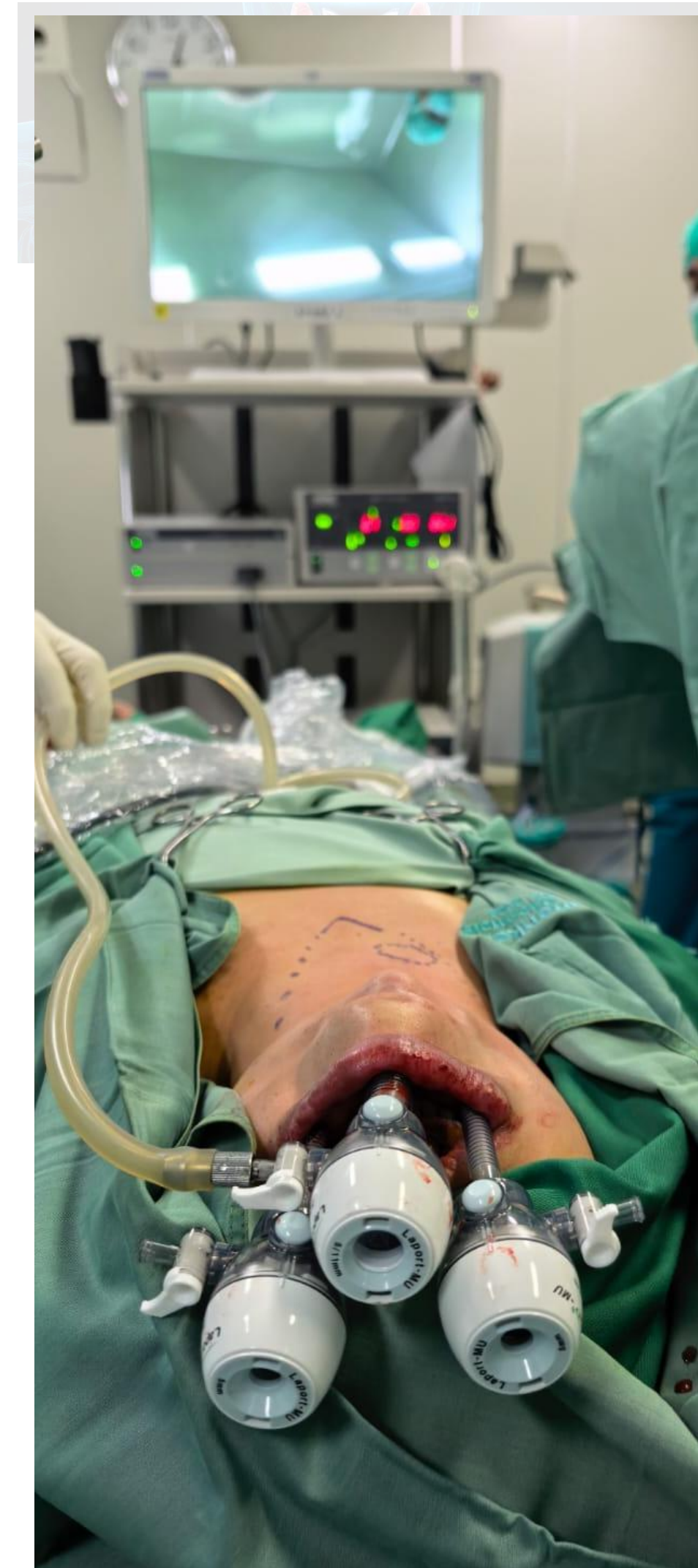
- Surgery : Endoscopic or Robotic (Low Risk) & Open (High Risk) Thyroidectomy, RND
- Internal Radiation / Radioablation with I131
- Medication : Levothyroxin 100 – 300 mcgr
- Targeted Therapy : Lenvatinib



# ENDOSCOPIC THYROIDECTOMY



**TOETVA**



► JAMA Surg. 2017 Sep 6;153(1):21–27. doi: [10.1001/jamasurg.2017.3366](https://doi.org/10.1001/jamasurg.2017.3366) 

## Safety and Outcomes of the Transoral Endoscopic Thyroidectomy Vestibular Approach

[Angkoon Anuwong](#)<sup>1,✉</sup>, [Khwannara Ketwong](#)<sup>1</sup>, [Pornpeera Jitpratoom](#)<sup>1</sup>, [Thanyawat Sasanakietkul](#)<sup>1</sup>, [Quan-Yang Duh](#)<sup>2</sup>

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PMCID: PMC5833624 PMID: [28877292](https://pubmed.ncbi.nlm.nih.gov/28877292/)



# TORT

Trans Oral  
Robotic Thyroidectomy

MANAGEMENT OF THYROID NODULES :  
HOW TO MAKE A GOOD DIAGNOSIS ?

Live Webinar, September 19<sup>th</sup> 2026



# TORT

Trans Oral  
Robotic Thyroidectomy

MANAGEMENT OF THYROID NODULES :  
HOW TO MAKE A GOOD DIAGNOSIS ?

Live Webinar, September 19<sup>th</sup> 2026



# Poor Differentiated Carcinoma Thyroid

- Surgery : Total Thyroidectomy, debulking
- External Radiation
- Chemoterapy
- Targeted therapy



| Drug                                                                         | Target                                                                 | Clinical.gov ID | Phase   |
|------------------------------------------------------------------------------|------------------------------------------------------------------------|-----------------|---------|
| Lenvatinib                                                                   | Multi-Targeted ( <i>i.e.</i> , VEGFR, FGFR, PDGFR $\alpha$ , RET, KIT) | NCT02726503     | II      |
| Trametinib + Paclitaxel                                                      | MEK, BRAF                                                              | NCT03085056     | Early I |
| Durvalumab + Tremelimumab + Stereotactic Body Radiotherapy                   | PD-1 and CTL-4, respectively                                           | NCT03122496     | I       |
| MLN0128                                                                      | mTOR                                                                   | NCT02244463     | II      |
| Pembrolizumab                                                                | PD-1                                                                   | NCT02688608     | II      |
| Nexavar                                                                      | Multi-Targeted ( <i>i.e.</i> , RAF, MEK, ERK, VEGFR, PDGFR)            | NCT03565536     | II      |
| Efatutazone + Paclitaxel                                                     | PPAR $\gamma$                                                          | NCT02152137     | II      |
| Intensity-Modulated Radiation Therapy + Paclitaxel + Pazopanib Hydrochloride | Multi-Targeted ( <i>i.e.</i> , VEGFR, PDGFR, c-kit)                    | NCT01236547     | II      |
| Atezolizumab + Bevacizumab + Cobimetinib + Paclitaxel + Vemurafenib          | PD-1, VEGFR, MEK and BRAF, respectively                                | NCT03181100     | II      |

If you fail, try  
again once more.  
And if you fail  
again, try 10  
times more.

Bob Andinata

Thank  
you!

1<sup>st</sup> TORT in Indonesia : March 30<sup>th</sup> 2026

